

TOZEUR NEFTA, Tunisia

WMO: 607600

Lat: 33.940N

Lon: 8.111E

Elev: 287

StdP: 14.54

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	40.8	42.9	19.8	15.0	61.4	23.1	17.6	59.4	28.2	59.4	24.8	58.5	6.8	270	0.534

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.4	111.2	69.9	107.8	69.8	104.4	69.7	76.5	92.6	75.0	91.3	73.8	91.2	8.8	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.1	119.9	83.3	70.2	112.3	82.8	68.4	105.4	82.5	40.0	92.1	38.6	91.4	37.5	91.0	85.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.7	24.9	22.4	36.0	116.0	2.3	2.2	34.4	117.6	33.1	118.9	31.8	120.1	30.2	121.6		
			32.3	78.6	2.5	2.7	30.5	80.6	29.1	82.1	27.7	83.6	25.9	85.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	72.8	53.7	56.7	63.5	70.5	78.8	86.8	92.0	91.6	84.5	75.6	63.4	55.4	(6)
(7)		DBStd	14.57	3.70	4.99	5.98	6.03	6.19	5.99	4.77	4.39	5.77	5.65	5.47	4.15	(7)
(8)		HDD50	22	11	5	0	0	0	0	0	0	0	0	0	6	(8)
(9)		HDD65	1095	350	237	99	16	0	0	0	0	0	2	93	298	(9)
(10)		CDD50	8344	126	193	419	614	891	1105	1302	1291	1034	792	403	174	(10)
(11)		CDD65	3944	0	5	54	180	427	655	837	826	584	329	46	1	(11)
(12)		CDH74	52631	1	35	456	1621	4822	9252	13160	12811	7447	2818	205	3	(12)
(13)		CDH80	31479	0	4	122	606	2473	5753	8871	8457	4085	1081	27	0	(13)
(14)	Wind	WSAvg	10.5	8.5	9.3	10.9	12.2	13.0	13.0	12.5	11.6	10.8	8.7	8.0	7.8	(14)
(15)	Precipitation	PrecAvg	4.6	0.7	0.3	0.4	0.7	0.2	0.1	0.0	0.1	0.5	0.4	0.5	0.5	(15)
(16)		PrecMax	12.6	5.4	2.4	1.3	3.8	1.6	2.2	0.3	1.1	3.7	2.0	3.1	3.5	(16)
(17)		PrecMin	1.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	2.8	1.3	0.5	0.4	1.0	0.4	0.4	0.1	0.3	0.7	0.5	0.7	0.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.6	78.6	89.6	97.3	106.2	113.0	115.2	113.3	107.8	97.2	83.0	72.0	(19)
(20)		0.4%	MCWB	54.2	55.8	59.6	62.4	66.2	69.0	71.0	70.1	69.5	65.9	61.1	55.1	(20)
(21)		2%	DB	66.4	73.4	82.8	91.1	100.1	108.1	111.6	109.7	102.6	93.3	79.0	68.4	(21)
(22)		2%	MCWB	52.5	54.9	58.2	61.6	65.8	68.4	70.3	70.7	69.4	66.2	60.7	54.7	(22)
(23)		5%	DB	64.4	69.8	78.9	86.3	95.4	104.0	108.0	106.2	98.9	89.5	75.8	66.3	(23)
(24)		5%	MCWB	51.8	53.2	57.2	60.7	65.1	68.1	69.9	70.9	70.0	66.0	59.6	54.0	(24)
(25)		10%	DB	62.3	66.5	75.2	82.6	91.6	100.1	104.3	103.5	95.3	85.9	73.3	64.1	(25)
(26)		10%	MCWB	51.0	51.9	56.0	59.6	64.2	67.7	69.8	70.9	69.9	65.7	58.4	53.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.6	59.3	63.9	66.5	70.8	74.8	77.9	79.0	77.8	73.8	66.2	59.6	(27)
(28)		0.4%	MCDWB	64.1	72.7	74.8	82.3	92.6	97.6	96.1	95.9	89.8	83.6	75.4	66.9	(28)
(29)		2%	WB	55.3	56.9	60.8	64.5	68.7	72.4	75.3	76.9	75.6	71.7	63.8	57.2	(29)
(30)		2%	MCDWB	62.0	70.4	76.7	80.9	90.5	95.0	95.5	93.6	87.9	82.3	73.6	64.5	(30)
(31)		5%	WB	53.6	54.9	58.9	62.9	67.3	70.7	73.8	75.3	74.3	69.9	61.9	55.6	(31)
(32)		5%	MCDWB	60.9	66.2	75.2	80.0	89.0	94.1	95.5	91.8	87.1	81.1	72.3	63.2	(32)
(33)		10%	WB	52.0	53.1	57.3	61.5	66.0	69.4	72.5	74.0	72.9	68.5	60.0	54.1	(33)
(34)		10%	MCDWB	59.9	63.6	72.0	78.0	86.9	92.9	94.7	91.9	86.5	80.1	70.5	62.1	(34)
(35)	Mean Daily Temperature Range	MDBR	17.4	18.8	20.1	21.1	22.1	23.0	23.4	21.9	19.3	18.3	17.4	16.7	(35)	
(36)		5% DB	MCDBR	19.4	22.1	24.4	26.0	26.2	26.8	27.5	25.3	22.7	21.7	19.0	18.5	(36)
(37)		5% DB	MCWBR	10.0	10.4	9.8	9.1	8.3	7.6	8.0	7.8	7.6	8.1	8.2	9.4	(37)
(38)		5% WB	MCDBR	16.7	19.4	21.9	22.6	23.3	24.1	24.3	22.1	19.8	17.8	17.1	16.4	(38)
(39)		MCWBR	9.5	9.8	9.6	8.9	8.8	8.6	8.4	8.1	7.9	7.8	8.8	9.1	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.359	0.387	0.480	0.534	0.572	0.547	0.579	0.559	0.542	0.487	0.419	0.371	(40)	
(41)		taud	2.223	2.122	1.859	1.753	1.698	1.766	1.713	1.763	1.813	1.917	2.052	2.179	(41)	
(42)		Ebn at Noon	264	268	251	244	236	241	233	234	230	231	237	250	(42)	
(43)		Edh at Noon	35	43	61	71	76	71	75	70	63	53	42	35	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1033	1348	1706	2058	2278	2481	2472	2205	1813	1422	1090	922	(44)	
(45)		RadStd	56	62	73	82	100	79	56	65	61	60	41	55	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	-2.05	N/A	N/A	N/A	N/A	N/A	+193	+202	(46)
(47)	Regional (6 neighbors)	N/A	N/A	-1.28	+1.19	N/A	N/A	N/A	N/A	+171	+149	(47)

Nomenclature: See separate page